TEXORA · LECTURE SHEET

Safety in Textile Mill

A textile mill concentrates rotating machinery, flammable fibre and dust, steam, chemicals, electricity and heavy manual work in one building. Safety is the management system that keeps those hazards from becoming injuries. This sheet covers accidents and their causes, the hazard map of each department, safety rules, first aid, how safety thinking evolved, and risk management with a worked risk-matrix assessment.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 11 · LEVEL 4 TERM 1

1. Accidents — definition and types

An **accident** is an unplanned event that causes (or could cause) injury, ill health, damage or loss. A **near miss** is an accident without injury — the most valuable free lesson a mill gets, if it is reported. Accidents are classified:

- **By severity:** fatal; lost-time (major) injury; minor injury (first-aid case); near miss; property-damage only.
- **By mechanism:** contact with moving machinery (entanglement, cutting, crushing); falls (same level — slips/trips; from height); struck by falling/flying objects; manual-handling injuries; burns and scalds (steam, hot surfaces, chemicals); electric shock; fire and explosion; exposure to dust, noise and chemicals (often slow, classed as occupational ill health rather than single accidents).

2. Causes and effects

Accident investigation consistently finds layered causes — blaming only the injured worker explains nothing and prevents nothing:

- **Unsafe acts:** removing or defeating guards, cleaning moving machinery, wrong lifting, rushing, unauthorised work on live equipment.
- **Unsafe conditions:** missing guards, poor housekeeping and blocked gangways, defective tools, poor lighting (Topic 10), oil on floors, unlabelled chemicals.
- **Root (management) causes:** no training or supervision, production pressure overriding permits, maintenance backlog (Topic 9), safety rules existing on paper only.

Effects: human — pain, disability, death, family income loss; legal — compensation, prosecution, closure; commercial — lost production, machine damage, insurance and audit consequences (buyer

compliance audits treat safety findings as critical), morale and reputation damage. The direct insured cost is typically only the visible part; uninsured losses (downtime, retraining, damage) are larger.

3. Industrial hazards by section

SECTION	PRINCIPAL HAZARDS	TYPICAL CONTROLS
Spinning (blowroom, carding, ring)	Fire in opened fibre; dust; rotating rollers and beaters; noise; fly accumulation	Metal detection and spark/fire protection, dust extraction (LEV), guarded nip points, hearing protection zones, housekeeping
Weaving	Shuttle/sley injuries on older looms; warp-line cuts; noise (the loudest common department); falling beams; manual handling	Modern looms without flying shuttles where possible, guards, beam-handling aids, noise control and PPE, gangway discipline
Wet processing (dyeing, finishing)	Steam and hot liquor scalds; chemical burns/splashes (alkali, acid, dyes); slippery floors; pressure vessels; stenter heat and fumes	Insulation and guarding of hot lines, chemical PPE and eyewash/showers, bunded chemical stores with labels/SDS, drainage and anti-slip floors, boiler discipline (Topic 3)
Apparel (cutting, sewing)	Band-knife and scissors cuts; needle punctures; eye injuries from broken needles; ergonomic strain (Topic 10)	Knife guards and mesh gloves, needle guards and eye protection, broken-needle procedure, ergonomic workstations
Utilities & warehouse (all sections)	Boiler explosion risk; electrical work; forklift–pedestrian collision (Topic 5); bale/stack collapse; fire load of stored fibre and cloth	Boiler mountings and inspection, lockout–tagout, traffic separation, stacking limits, fire detection, hydrants and drills

Fire deserves emphasis: fibre, dust, oil and packaging make textiles a high fire-load industry. Prevention is housekeeping (fly and waste removal), ignition control (electrical maintenance, hot-work permits, no smoking), detection and alarm, compartmentation, and practised evacuation — exits never locked or blocked during working hours.

4. Safety rules (mill standing rules)

1. Never clean, oil or adjust moving machinery; stop, isolate and lock out first (guards refitted before restart).
2. Wear the PPE issued for the section (hearing, eye, gloves, apron, mask) — and no loose clothing, ties, ornaments or untied long hair near rotating parts.
3. Keep gangways and exits clear; stack within marked limits; clean spills and fly at once.

4. Use lifting aids for heavy loads; never ride on forklifts or stand under suspended loads.
5. Handle chemicals only with label, SDS and PPE knowledge; never mix unlabelled containers.
6. Report every injury, near miss and damaged guard — reporting is protected, concealment is a disciplinary matter.
7. Know the alarm, your exit route and assembly point; take part in drills.
8. Only authorised, trained persons operate boilers, electrical switchgear, cranes and forklifts.

5. First aid

First aid is the immediate care given before medical help takes over; its aims are to preserve life, prevent worsening and promote recovery. Mill provision: trained first-aiders on every shift, stocked first-aid boxes at marked points, eyewash stations and safety showers in chemical areas, a stretcher and a clear vehicle route, and the emergency numbers displayed. Key responses:

- **Bleeding:** direct pressure with sterile dressing; raise the limb; do not remove embedded objects.
- **Burns/scalds:** cool under running water (about 20 minutes in current guidance), cover with sterile non-stick dressing; never apply pastes or peel off stuck clothing.
- **Chemical splash to eyes:** irrigate at the eyewash immediately and continuously, then hospital.
- **Electric shock:** isolate the supply before touching the casualty; then airway/breathing checks and CPR if trained.
- **Fractures:** immobilise as found; move only if in danger.
- **Amputation/entanglement:** stop the machine, control bleeding, preserve the severed part cool and clean for hospital, urgent transfer.

Every first-aid treatment is recorded — the record is accident data for §7.

6. Evolution of modern safety concepts

- **Compensation era (early industrial):** safety meant paying after injury; prevention was minimal — the era of unguarded line shafts.
- **Legislation and inspection:** factory acts imposed guarding, hours and inspectorates; in Bangladesh the Labour Act and factory rules, with the Department of Inspection for Factories and Establishments, carry this role today.
- **Technical/engineering era:** guards, interlocks, ventilation and machine design standards (Topics 6, 10) engineered hazards out.
- **Management-systems era:** safety as a managed system — policy, risk assessment, training, audit — codified in occupational health and safety management standards (ISO 45001 line) and, in the

garment sector, buyer and accord-style building/fire/electrical safety programmes that transformed factory safety practice in Bangladesh after 2013.

- **Behavioural and culture era (current):** near-miss reporting, safety culture, worker participation and leading indicators (inspections done, hazards closed) rather than only counting injuries after the fact.

7. Safety risk management techniques

Risk combines how likely harm is with how severe it would be: Risk = Likelihood × Severity. The management cycle is: **identify hazards → assess risk → control (eliminate, substitute, engineer, administrate, PPE — in that hierarchy order) → monitor and review**. Techniques include workplace inspections and checklists, job safety analysis (breaking a job into steps and hazard-checking each), permit-to-work for hot work/confined spaces/electrical work, accident and near-miss investigation, safety audits, and the **risk matrix** below for prioritising.

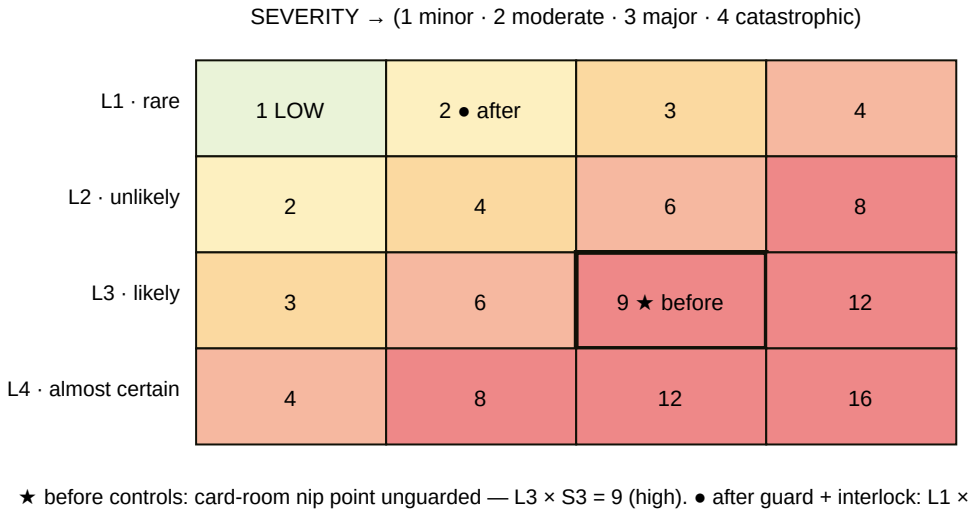


Figure 1 — 4 × 4 risk matrix (score = likelihood × severity; cell values shown). The worked example below moves one hazard from 9 (high, red-bordered cell) to 2.

WORKED EXAMPLE — RISK ASSESSMENT OF A CARDING-MACHINE NIP POINT

Hazard: operator's hand drawn into an unguarded roller nip point while clearing lap-ups on a running card.

Before controls: likelihood 3 (likely — the task is frequent and the guard is missing) × severity 3 (major — crush/amputation injury possible) = **risk 9, HIGH**. Action: stop — interim ban on clearing while running.

Controls (hierarchy order): engineering — fit the nip guard with an interlock so rollers stop when it opens; administrative — written clearing procedure (machine stopped, isolated), training and supervision; PPE — close-fitting clothing rule (loose items banned near rollers).

> **After controls:** likelihood 1 (rare — interlock prevents running with guard open) × severity 2 (moderate — residual contact risk during stopped clearing) = **risk 2, LOW**. Residual risk accepted with monitoring: interlock tested weekly (Topic 9 schedule), near misses reviewed monthly.

Glossary

Hazard — anything with potential to cause harm. **Risk** — likelihood × severity of harm from a hazard. **Near miss** — an incident that could have caused harm but did not. **Lockout–tagout** — isolation of energy with a personal lock and warning tag before work on machinery. **PPE** — personal protective equipment, the last line of control. **SDS** — safety data sheet for a chemical. **Permit to work** — written authorisation for high-risk tasks. **Hierarchy of controls** — elimination → substitution → engineering → administration → PPE, in order of reliability.

Key points

• Causes are layered: unsafe acts and conditions grow from root management causes — investigate to the root.
• Each section has its own hazard signature: fire/dust in spinning, noise in weaving, chemicals/steam in wet processing, knives/needles in apparel.
• First aid priorities: cool burns, irrigate eyes, pressure on bleeding, isolate power before touching a shock casualty — and record everything.
• Risk = likelihood × severity; controls follow the hierarchy; worked example: 3 × 3 = 9 (high) → 1 × 2 = 2 (low) after guarding and interlock.

Practice

1. Score this hazard before and after the stated control, using the 4 × 4 matrix: forklifts crossing a pedestrian gangway (before: L3, S4). Control: separated, barriered pedestrian route with marked crossings (after: L1, S4). State both scores and bands, and explain why severity does not change.
2. For a stenter operator's heat/fume exposure, give one control at each level of the hierarchy.
3. A worker scalds a forearm on an unlagged steam line. Write the first-aid steps in order, and name the root-cause questions the investigation must ask.
4. Why is a rising number of near-miss reports usually a sign of a *healthier* safety culture, not a worsening one?

Answer key

1. Before: 3 × 4 = **12, high**. After: 1 × 4 = **4, medium-low on this matrix**. Severity stays 4 because if a collision still occurred the outcome would be equally serious — the barrier reduces only the likelihood of contact.
2. Elimination: remove the need to stand at the hot delivery end (automate batch handling). Substitution: lower-temperature process chemistry where quality allows. Engineering: enclosure + LEV exhaust over the stenter. Administrative: exposure-time limits, rotation, training. PPE: heat gloves, apron, eye protection for close tasks.
3. Steps: move away from the line, cool under running water about 20 minutes, remove jewellery/loose items around the burn, cover with sterile non-stick dressing, seek medical assessment, record the treatment. Root-cause questions: why was the line unlagged (insulation removed/not replaced after maintenance?), why was the route beside it, what inspection should have caught it, was it reported before?
4. Near misses were always occurring; reporting them means workers trust the system. Each report is a free correction opportunity — hazards get controlled before the same event produces an injury. The trend to fear is zero reports in a hazardous mill, which signals concealment, not safety.
